A Terminal-cell branch sets
2-branch, 5,000 cells
3,684
2-branch, 2,000 cells
1,470
simulated branch
1,357
2-branch, 500 cells
366
0
1000
2000
3000
4000
Aligned terminal cells
B One-to-one branch F1
Delta 0.30 [-0.01, 0.60]; P = 0.250
2-branch, 5,000 cells
2-branch, 2,000 cells
simulated branch
2-branch, 500 cells
0.00
0.25
0.50
0.75
1.00
F1
C Lineage ARI
Delta 0.70 [0.29, 1.00]; P = 0.125
2-branch, 5,000 cells
2-branch, 2,000 cells
simulated branch
2-branch, 500 cells
0.00
0.25
0.50
0.75
1.00
Adjusted Rand index
slingshot
LineaLinkR
D Three- and four-branch ordering stress
TSCAN
MultistageOT
LineaLinkR
scTour
scFates
Palantir
slingshot
monocle3
SCORPIUS
pCreode
DPT
StaVia
3br 2k
3br 2k batch
3br 2k drop30
3br 2k drop50
3br 2k drop70
3br 2k drop90
3br 500
3br 5k
4br 2k
4br 500
4br 5k
Spearman
-1.0
-0.5
0.0
0.5
1.0
E Ordering accuracy across 19 datasets
TSCAN
MultistageOT
LineaLinkR
scTour
scFates
monocle3
SCORPIUS
slingshot
Palantir
pCreode
DPT
StaVia
0.0
0.2
0.4
0.6
0.8
Mean Spearman rho (95% bootstrap CI)
F Output availability
LineaLinkR
scTour
SCORPIUS
TSCAN
StaVia
slingshot
scFates
pCreode
Palantir
MultistageOT
monocle3
DPT
0
6
12
18
24
Datasets with complete output
G Fork-evidence recovery
600 module-level replicates; AUROC = 0.975
nested fork
100%
focal branch
100%
dropout
74%
shared trend
0%
low expression spike
0%
donor confounded null
0%
0%
25%
50%
75%
100%
Detection rate
Expected fork
Null / rejected